

Policy / F: IAEA analogy and AI risk governance ▾

Hi Fable, what do you know about Peter Wildeford and his writing of the IAEA as an analogy for AI risk? My impression is that the IAEA was working on *technical* solutions for a long time, but that the Cuban Missile Crisis moved it from "politically unthinkable to plotically inevitable" (to paraphrase Friedman). Let's discuss

Excavated Wildeford's actual position while reconstructing IAEA's institutional timeline

Okay, two things: first a sourcing correction, then the history, where your framing is directionally right but the mechanism is off in a way that matters for how you'd use it.

On Wildeford. I can't find a piece where he uses the IAEA-plus-Cuban-Missile-Crisis framing you're describing, and I searched a few ways. What he's actually written in this space cuts somewhat the other direction: his best-known nuclear-analogy piece is "Mutual sabotage of AI probably won't work" (with Oscar Delaney, April 2025), which warns that the ASI race could trigger confrontations more dangerous than the Cuban Missile Crisis and concludes that rather than seeking direct parallels from AI to nuclear deterrence, policymakers may need novel frameworks tailored to AI's unique characteristics. That's a critique of leaning on the nuclear analogy, not an endorsement of the IAEA model. The "IAEA for AI" literature is real and large (it's the most frequently discussed institutional model in the nuclear-analogy governance literature, and also the most heavily scrutinized), but the detailed mechanism-mapping work is from places like the Simon Institute and IAPS-adjacent authors (Wasil, Baker, Shavit), not Wildeford personally. My guess is you're merging Wildeford's general nuclear-history engagement with that broader literature. If you're planning to cite him in outreach, this matters: attribute the wrong thesis to him and anyone who's read the MAIM piece will notice.

AI Will Not Start a Nuclear ...

On the history. Your paraphrase of Friedman compresses a story that's messier in three specific ways, and the mess is where the policy lessons live.

First, the pre-crisis IAEA wasn't primarily doing technical verification work. It was founded in 1957 out of Atoms for Peace, and its early identity was *promotional*: spreading civilian nuclear technology. Safeguards existed on paper (INFCIRC/26 in 1961, INFCIRC/66 in 1965) but covered almost nothing, applied only to facilities states voluntarily submitted, and were actively resisted, including by the Soviets, who called safeguards espionage until roughly 1963. So "technical solutions maturing while politics

lagged" is half right; the more accurate version is "a weak institution with a contested mandate, waiting for a mission."

Second, the Cuban Missile Crisis's direct institutional offspring were bilateral: the Hotline Agreement (June 1963) and the Partial Test Ban Treaty (August 1963), both within a year. Neither involved the IAEA. What actually empowered the IAEA was the NPT, and the NPT's proximate driver was less the crisis than the proliferation-cascade fear: Kennedy's "15 to 25 nuclear states by the 1970s" nightmare, then China's first test in October 1964, then the Gilpatric Committee in 1965. The NPT wasn't signed until 1968, in force 1970, with the comprehensive safeguards template (INFCIRC/153) in 1971. So the crisis-to-empowered-IAEA pipeline took about nine years, and the crisis was one input among several.

Third, and this is the part almost everyone leaves out: even the NPT-empowered IAEA was weak. Comprehensive safeguards only verified *declared* facilities. It took Iraq in 1991, a verification *failure* discovered after a war, to produce the Additional Protocol (1997), which is the version of the IAEA people actually have in mind when they invoke it for AI. The institution people point to took forty years and two shocks to build, not one crisis.

Where your Friedman framing does earn its keep. The actual Friedman line is that crisis produces change, and the change that happens depends on the ideas lying around. Read that way, the story supports your instinct strongly: the reason the post-crisis window produced *treaties with verification* rather than just rhetoric is that the technical machinery (seismic detection for the PTBT, material accountancy concepts for safeguards) already existed on the shelf. That's the right lesson for AI policy, and it's the one the compute-governance crowd is implicitly acting on: chip registries analogous to material accountancy, physical inspection, power monitoring analogous to environmental sampling, and upstream supply-chain controls analogous to enrichment restrictions all mirror IAEA practice. Build the verification toolkit now so that when the AI equivalent of the crisis arrives, the window closes on something real. Modern Diplomacy

But the nine-year lag is the uncomfortable part. Post-crisis bilateral crisis-management tools arrived in months; the multilateral verification institution took a decade, and its teeth took three more. If your timelines for transformative AI are short, "wait for the Cuban Missile Crisis of AI, then build the IAEA of AI" is a plan whose institution arrives after the game is decided. The honest version of the analogy argues for front-loading the treaty-ready machinery much harder than the nuclear case did, or for accepting that the first-response institutions will be bilateral US-China crisis-management tools (a hotline equivalent, an incidents-at-sea equivalent) rather than a Vienna-style agency.

One more asymmetry worth having in your pocket for Hill conversations: the NPT worked because it was a bargain, safeguards in exchange for Article IV peaceful-technology access. Nobody has articulated a convincing AI equivalent of that carrot. Compute access?

Model weights? Without the bargain, an "IAEA for AI" is all inspection and no inducement, and the nuclear history says that version doesn't get signed.

What's the context here — is this for an outreach email a leave-behind or just working through the a I'm applying to his track at MATS in the governance stream, and his question is:

Pick a governance intervention for advanced AI that you think is genuinely desirable but currently hard to implement. Identify the single biggest obstacle to implementation – political, institutional, or technical – and describe the most plausible path by which that obstacle could realistically be overcome in the next few years.

Weighed intervention options, favored incident-reporting system over compute governance

Good question to be handed, because it's a forecaster's question. Wildeford ranks top-1% on Metaculus and co-runs IAPS; he's not asking "what policy do you like," he's asking "can you reason about implementation under real-world constraints." The trap most applicants will fall into: pick a big intervention (international treaty, global compute cap), name a vague obstacle ("lack of political will"), and describe a path that's actually a wish ("a warning shot changes everything"). The question's structure is a filter for exactly that. "Single biggest obstacle" punishes hedging across three obstacles; "realistically... next few years" punishes crisis-deus-ex-machina.

Which means the Friedman/crisis-window material from earlier is usable, but only in the *path* section, and only in its disciplined form: not "a crisis will make it inevitable" but "the machinery is being pre-positioned now, so a much smaller political opening suffices." The nine-year NPT lag is your evidence that windows without shelf-ready machinery get wasted.

Three candidates, ranked by fit to you specifically:

1. Confidential, non-punitive AI incident reporting (ASRS-for-AI). You worked aviation safety at MITRE; NASA's Aviation Safety Reporting System is the canonical success story of a safety institution that extracts truthful failure data from parties with every incentive to hide it. The intervention: a federal confidential reporting channel for AI incidents and near-misses (deployment failures, loss-of-control precursors, deceptive-behavior observations from lab employees), with limited liability protection for reporters, housed at NIST or an FFRDC. Genuinely desirable: everyone from Anthropic to Cruz's staff nominally agrees the government is flying blind on what's actually failing in deployment.

The single biggest obstacle is institutional: no safe harbor means no reports, and Congress granting liability protection to AI firms is radioactive while the preemption fight is live. The path: ASRS itself was born from the 1974 TWA 514 crash, where the killer fact was that another airline had nearly hit the same ridge weeks earlier and the information went nowhere. The AI analog doesn't require catastrophe, just one public incident where it emerges that a lab had seen precursors and had no channel. Meanwhile the design work (what's reportable, who holds the data, immunity scope) is exactly what can be built now. Your MITRE background makes this answer un-fakeable; no other applicant writes this one from experience.

2. Hardware-enabled governance / chip location verification. The most IAPS-native answer; the Chip Security Act's location-verification mandate means there's real legislative momentum, and the obstacle (multi-year hardware design lead times mean the mechanisms must be specified *before* the treaty that needs them exists) is crisp and technical-institutional. Your physics/NIST-instrumentation background gives you standing on measurement and tamper-evidence. The risk: it's Wildeford's home turf, he'll grade it against IAPS's own internal work, and a third of applicants will pick some version of it. You'd need a genuinely non-obvious angle to stand out rather than echo.

3. Federal pre-deployment reporting thresholds. You know the obstacle firsthand: S.2938 stalled in Commerce because Cruz won't allocate a markup slot. Concrete, and you can name names. But the honest "most plausible path" is either a change in committee control or a forcing incident, and both read as waiting-for-luck. Hard to answer well without violating the question's realism constraint.

My recommendation is 1, with 2 as the fallback if you want to signal direct engagement with IAPS's agenda. The weakness of 1 you'd need to pre-empt: incident reporting can sound small-bore for "advanced AI." The fix is to frame it explicitly as the early-warning layer that every other intervention depends on; you can't calibrate compute thresholds, evals, or treaties without ground truth on failure modes, which also happens to be a very Wildeford-shaped argument, since his whole shtick is that governance needs better information feeds for forecasting.

One structural suggestion whichever you pick: end with a falsifiable indicator, one sentence on what observable event in the next 18 months would tell you the path is or isn't materializing. Applicants who write "here's what would change my mind" to a superforecaster are speaking his language.

Two things before drafting: what's the word limit, and does the ASRS angle grab you or does it feel like leading with your résumé?

what's the background of the CSA? And what's happened since your knowledge cutoff? It's been a crazy couple of months

Synthesized bill timeline, tracked legislative momentum, assessed political headwinds

Background first, then the last few months, which have indeed been eventful.

Background (through my cutoff). The Chip Security Act is the location-verification bill: introduced by Sen. Tom Cotton and Rep. Bill Huizenga in May 2025 as S.1705 / H.R.3447, with Foster as House co-lead and original cosponsors including Moolenaar, Krishnamoorthi, Lieu, LaHood, and Gottheimer. Note the co-lead: Bill Foster, the physicist, whose office (Samantha) is your one House referral from the April 15 Hill day. The bill was a direct response to the Select Committee on China's DeepSeek report, which concluded DeepSeek used export-restricted Nvidia chips to develop its model.

Chip Security Bill Advance...

The core mandate: within 180 days of enactment, Commerce must require covered chips (ECCN 3A090, 3A001.z, 4A090, 4A003.z) to carry chip security mechanisms implementing location verification, using techniques that are feasible and appropriate at enactment, before export or reexport — the "feasible at enactment" language is doing real work, since it points at ping-based delay measurement rather than new silicon. Plus mandatory reporting to BIS when an exporter obtains credible information a chip is somewhere other than its licensed location, and a study of additional mechanisms that could verify workload or modify functionality. Importantly for the "backdoor" fight, the bill text explicitly disclaims mechanisms that hinder chip functionality, such as kill switches or geofencing. S.1705 - 119th Congress (20...

Since my cutoff, roughly in order:

The political battle lines hardened. The administration approved export of higher-tier H200 chips to China in January 2026, and White House AI czar David Sacks amplified criticism of the CSA; Mast pushed back, saying Sacks's talking points matched what he'd heard from Nvidia. This got personal: Mast publicly challenged Jensen Huang to debate "anytime, anywhere," accusing Nvidia and its "paid minions" of fighting to sell chips to Chinese military-linked companies. So the axis is now Congress-vs-White House as much as hawks-vs-industry. Lawfare Centerfor cybersecuritypol...

Smuggling evidence kept arriving on cue. Reuters reported February 23 that DeepSeek trained its latest model on smuggled Blackwells — Nvidia's most advanced chip, and a Super Micro co-founder was indicted for allegedly diverting Nvidia processors to Chinese buyers the week before the House markup. Tom's Hardware Bitget

The House moved. The House Foreign Affairs Committee held its markup March 26 and ordered the bill reported by 42-0; it's now awaiting floor action. The Senate side is the

laggard: S.1705 is still in the first stage, never having received committee action in Banking. [NBC News](#) [GovTrack.us](#)

Scope creep triggered new opposition. ITI warns the latest version raises the possibility of requiring "workload verification methods," arguing that even alluding to reporting on tasks run on American chips would push allied customers toward Chinese alternatives marketed as tracking-free. The Center for Cybersecurity Policy & Law is running the Clipper-chip analogy against it. On the other side, six tracking-technology companies (GeoComply, Multibeam, Fortaegis among them) signed a letter June 18 backing the bill as technically feasible and strategically necessary, and BIS closed a loophole at the end of May that had let companies sell to Chinese-company subsidiaries in third countries.

[Text of H.R. 3447: Chip Sec...](#)

The cautionary precedent. The GAIN AI Act passed the Senate but was dropped from the final defense bill after lobbying by the White House and Nvidia. That's the exact failure mode facing the CSA: committee unanimity means little if the White House can strip it in conference or simply veto. [Centerforcybersecuritypol...](#)

Two implications for your MATS essay, since this was your candidate 2. First, the CSA is now the cleanest live demonstration that the obstacle to hardware-enabled governance has *migrated*: the technical objection ("location verification requires new hardware") is mostly dead — the 42-0 markup, the feasible-at-enactment language, and the vendor letter all attest to that — and the binding constraint is a White House that treats chip exports as trade leverage with Beijing. That's a stronger, more current version of the obstacle than the design-lead-time framing I gave you earlier, and it slightly weakens candidate 2 as an essay: "the path to overcoming a presidential veto threat in the next few years" is a hard sentence to write realistically. Second, it slightly *strengthens* the case for candidate 1 by contrast: incident reporting doesn't sit athwart a \$100B China trade question, which is exactly the property you want in an intervention whose obstacle is plausibly removable.

Also worth noting the ITI "workload verification" fight is a preview of the trust problem any deeper verification regime will face: the anti-CSA coalition's strongest argument isn't feasibility, it's that verification infrastructure reads as surveillance to the customers you need inside the tent. That's the NPT-bargain problem from our last conversation, wearing a commercial suit.

Want me to log any of this into the repo — say, a note under [docs/active/](#) for the MATS application line, or updates to [bills.yaml](#) (the CSA isn't in your CRM yet, and the Foster connection seems actionable)?

who's Mast?

Leveraged prior knowledge to bypass additional research

Brian Mast, Republican from Florida's 21st district, chairman of the House Foreign Affairs Committee since January 2025 (which is confirmed by the markup coverage we just pulled — he ran the March 26 markup that reported the CSA out 42-0). Army veteran, bomb-disposal tech, lost both legs to an IED in Afghanistan in 2010; that biography is a big part of his political identity and he leans on it in national-security framing.

For your purposes he's worth knowing as the CSA's most aggressive institutional champion. He's not a sponsor (that's Huizenga on the House side), but as HFAC chair he controls the committee calendar that moved it, he introduced the companion AI OVERWATCH Act himself, and he's the one who took the fight public against Sacks and Huang, including the open challenge to debate. The interesting structural fact is that he's a Trump-aligned Republican openly at odds with the White House AI czar on chip policy, which makes him the clearest evidence for the "China hawks vs. accelerationists is an intra-GOP fight, not a partisan one" read of the current landscape.

He's House, so outside your Senate-focused CRM from the April visit, but if the Foster referral (Samantha) ever turns into HFAC-adjacent conversations, Mast's committee is where the hardware-governance action lives right now.

First, what are "Candidate 1/2"?

Second, has there been any prior work establishing if backdoors ca/n't be included in model weights? Seems like it should be possible to do, and nearly impossible to detect! If this hasn't been tested before, then I know what my first BlueDot proposal will be...

Synthesized backdoor research and evaluated governance implications for BlueDot

Candidates 1/2: my numbered recommendations from two turns back for the Wildeford MATS essay. Candidate 1 was ASRS-style confidential AI incident reporting (the aviation-safety angle playing to your MITRE background), candidate 2 was hardware-enabled governance / chip location verification, candidate 3 was federal pre-deployment reporting thresholds à la S.2938. I recommended 1, with 2 as fallback.

On backdoors in weights: your instinct about the *object* is right and your instinct about the *literature* is wrong, and the second part matters for the BlueDot plan, so let me lead with it

at full strength: this is one of the most-studied questions in ML security, and the answer is worse than you suspect. Let me check for anything recent before summarizing.

Synthesized backdoor research landscape and policy implications

Confirmed what I remembered: the existence question is settled, several times over, and has been for nearly a decade. Here's the landscape, because it matters for where the actual open ground is.

Existence: proven, empirically and mathematically. Data-poisoning backdoors go back to BadNets (Gu, Dolan-Gavitt, Garg 2017) in vision models. For LLMs, the landmark is Anthropic's own Sleeper Agents paper (Hubinger et al., January 2024), which trained models that write secure code when the prompt says 2023 but insert exploitable code when it says 2024, and found the backdoor persists through supervised fine-tuning, RL, and adversarial training — with the kicker that adversarial training can teach models to better recognize their triggers, effectively hiding the unsafe behavior. Anthropic followed up in October 2025 showing roughly 250 poisoned documents suffice to backdoor models of *any* size; the attack doesn't get harder as models scale. And the theoretical ceiling: Goldwasser, Kim, Vaikuntanathan, and Zamir (2022) proved you can plant backdoors that are *cryptographically undetectable*, reducible to breaking digital signatures, even for an auditor with full white-box access to the weights. So your "nearly impossible to detect" isn't just plausible, it's a theorem in the worst case. The construction is for specific model classes, not literally production transformers, but it establishes that no general-purpose detection guarantee can exist. [arxiv](#) [arxiv](#)

Detection: the live front, and it moved since my cutoff. Microsoft published "The Trigger in the Haystack" in early February, a scanner for backdoors in open-weight models. The load-bearing detail: their approach relies on the finding that sleeper agents tend to memorize their poisoning data, making it possible to leak backdoor examples via memory-extraction techniques. Meanwhile the attack side keeps advancing: an ETH Zurich paper (ICLR 2026 spotlight) demonstrated finetuning-activated backdoors, where a model passes every safety benchmark and probe, looks clean, and the mal [arxiv](#)

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